

CMPSC 202: Big-O Examples

Here are a few examples of proving Big-O bounds using the rules we introduced in Lecture 7. Feel free to try them before looking at the solution and post in Discord if you have any questions.

Example 1: Polynomial

Prove that $3n^2 + 2n + 1$ is $\mathcal{O}(n^2)$.

Proof. Our running time is a sum, so we can check each term individually, then combine the results using the rule that the sum of Big-O terms is dominated by the fastest growing term.

Looking at each summand, we have:

- $3n^2$ is $\mathcal{O}(n^2)$ by dropping constants.
- $2n$ is $\mathcal{O}(n)$ by dropping constants.
- 1 is $\mathcal{O}(1)$ by dropping constants.

Therefore, by the sum is max rule, we have:

$$3n^2 + 2n + 1 \text{ is } \mathcal{O}(n^2),$$

since n^2 dominates n and 1 as n grows large. □

Example 2: Logarithmic vs Polynomial

Prove that $T(n) = 4n \log(n)$ is $\mathcal{O}(n^2)$.

Proof. We want to show that $4n \log(n)$ is $\mathcal{O}(n^2)$. Let's think about we get from our starting point to the ending point.

First, by dropping constants, we have: $4n \log(n)$ is $\mathcal{O}(n \log(n))$.

Next, let's analyze the relationship between $n \log(n)$ and n^2 . Starting with each term of $n \log(n)$, we have:

- n is $\mathcal{O}(n)$ by dropping constants.
- $\log(n)$ is $\mathcal{O}(n)$ since $\log(n)$ grows slower than any polynomial function of n .

Therefore, by the multiplication is multiplication (also called the product rule) for Big-O, we have:

$$n \log(n) \text{ is } \mathcal{O}(n \cdot n) \text{ is } \mathcal{O}(n^2).$$

Combining this with the previous step by transitivity, we get:

$$4n \log(n) \text{ is } \mathcal{O}(n \log(n)) \text{ is } \mathcal{O}(n^2).$$

□

Note that this bound isn't "tight" because $4n \log(n)$ is actually $\mathcal{O}(n \log(n))$, which is a faster running time than $\mathcal{O}(n^2)$.

Example 3: Competing Terms

Prove that $T(n) = 5n^4 2^n + 3n^5$ is $\mathcal{O}(n^4 2^n)$.

Note: this one is tricky because we may likely do not know in advance which term will dominate for large n . We need to carefully analyze the growth rates of each term.

Proof. First, let's look at each term individually:

- $5n^4 2^n$ is $\mathcal{O}(n^4 2^n)$ by dropping constants.
- $3n^5$ is $\mathcal{O}(n^5)$ by dropping constants.

Next, we need to compare n^5 and $n^4 2^n$. First, let's look at $n^4 2^n$ using the product rule for Big-O. We have:

- n^4 is $\mathcal{O}(2^n)$ since 2^n grows faster than any polynomial function of n .
- 2^n is $\mathcal{O}(2^n)$ by reflexivity.

Therefore, by the multiplication is multiplication (also called the product rule) for Big-O, we have:

$$n^4 2^n \text{ is } \mathcal{O}(2^n \cdot 2^n) \text{ is } \mathcal{O}(2^{2n}).$$

Dropping constants, we can simplify $\mathcal{O}(2^{2n})$ to $\mathcal{O}(2^n)$. Therefore, $n^4 2^n$ is $\mathcal{O}(2^n)$.

Recall that exponential functions grow faster than polynomial functions. Therefore, $n^5 + n^4 2^n$ is $\mathcal{O}(2^n)$ by the sum is max rule (also called the summation rule). By transitivity, we have:

$$5n^4 2^n + 3n^5 \text{ is } \mathcal{O}(2^n).$$

□